Ann Mary Thomas

Data Analyst | Energy & Sustainability Analyst | AI/ML Researcher

annmaryttt@gmail.com | linkedin.com/in/ann-mary-thomas-6272aa200 | github.com/Ann-Mary-Thomas | Coventry, UK

PROFILE

Data, energy, and sustainability analyst combining mechanical engineering foundations with advanced data analytics and deep learning research. Currently at WMG, University of Warwick, embedded within 3T Additive Manufacturing as part of the £ 14.1M government-funded STRATA programme — building energy baselines, designing decarbonisation roadmaps, and delivering Power BI dashboards for Honeywell consortium partners. Research experience at London Metropolitan University's GENESIS Lab, where I co-authored two journal papers and designed TAL-Net — a Temporal Attention LSTM Network achieving 0.48% MAPE for electricity demand forecasting. Proven ability to translate complex datasets into clear, actionable insights for technical and non-technical stakeholders across industry, academia, and government bodies.

CORE SKILLS & TECHNOLOGIES

Data Analysis	Python (Pandas, NumPy, Matplotlib, scikit-learn, TensorFlow), SQL, Excel (Advanced), Statistical Analysis
AI & Machine Learning	LSTM, Bidirectional LSTM, Attention Mechanisms (TAL-Net), Time-Series Forecasting, SHAP Interpretability, Hybrid Deep Learning
Visualisation & BI	Power BI, Dashboard Design, KPI Reporting, Data Storytelling, HTML UI Development
Energy & Sustainability	Carbon Footprinting (Scope 1/2/3), CII/IMO Standards, ISO 50001/14001/14040/14044, IPMVP, Load Factor Modelling, ND-NEED 2022, CIBSE Guide F
Systems & Tools	Git, Jupyter Notebook, IoT Sub-metering, Data Pipeline Automation, Data Cleansing & Validation
Professional	Stakeholder Engagement, Technical Report Writing, Grant Reporting, Cross-team Collaboration, Research & Publication

PROFESSIONAL EXPERIENCE

Energy Consumption & Sustainability Analyst
Dec 2025 – Apr 2026

WMG, University of Warwick — 3T Additive Manufacturing (Project STRATA)

£14.1M ATI Programme / Innovate UK • Honeywell Consortium • National Centre for Connected Manufacturing

28,510 kWh	**6.65 tCO_2**	**£9,907/yr**	**15–18%**
Annual facility consumption	Annual carbon baseline	Energy cost equivalent	Projected annual reduction

Energy Baseline & Hotspot Analysis

- **Built complete energy performance baseline** for a 35-staff facility across 35+ equipment types — zero sub-meters, zero BMS; applied IPMVP Option A methodology (nameplate ratings, load factor modelling, site observation), validated against ND-NEED 2022, CIBSE Guide F, and Carbon Trust benchmarks (3.26 kWh/employee/day)
- Identified **HVAC (49%), water boiler (25%), and lighting (12%)** as primary consumption hotspots, directing intervention priorities

Scenario Design & Financial Modelling

- Designed **6 improvement scenarios with 5-year ROI analysis**: AC temperature optimisation (18% reduction), automatic shutdown compliance (16%), kitchen automation (59% sub-category) — combined **15–18% annual energy reduction**

Standards, Reporting & Digital Infrastructure

- Structured Energy & Sustainability Policy Framework to **ISO 50001, ISO 14001, ISO 14040/14044**; produced Digital Product Passport (DPP) equipment catalogue to LCA standards
- Specified IoT sub-metering infrastructure (sensor spec, vendor shortlist); participated in **ERP/MRP/MES integration** discussions for digital twin pipeline
- Built Power BI dashboards (kWh/employee, carbon intensity KPIs); presented findings to **WMG management, 3T senior leadership, and NCC/Honeywell consortium partners**

Energy Consumption & Sustainability Analyst (Intern)
Sep 2025 – Dec 2025

3T Additive Manufacturing — WMG Industry Placement

- **Achieved a 39.3% carbon footprint reduction** through systematic energy data analysis and targeted operational interventions across manufacturing processes
- Constructed facility energy performance baseline from zero — no existing sub-meters, BMS, or monitoring data; applied **load factor modelling** to nameplate ratings using IPMVP Option A methodology

- Designed and quantified **6 savings scenarios**: AC optimisation (18%), shutdown compliance (16%), kitchen automation (59% sub-category) — combined 15–18% projected annual reduction; produced 5-year financial projections and ROI analysis per scenario
- Aligned recommendations to **ISO 50001** (Energy Management Systems); designed sub-metering deployment roadmap to transition from estimated to real-time M&V baseline
- Built interactive Power BI dashboards for Scope 1/2 KPI monitoring; authored technical reports for management and funding bodies — directly leading to full analyst role at WMG

Research Assistant — Energy Forecasting & Carbon Analytics
Feb 2025 – Present

GENESIS Research Lab, London Metropolitan University (under Dr. Maitreyee Dey)

0.48% MAPE — electricity demand	**1.60%** MAPE — CO_2 emissions	**19×** Improvement over ARIMA	**$p < 0.0001$** Statistical significance

Model Architecture & Results
- Co-authored **2 peer-reviewed journal papers** on electricity demand and CO_2 emissions forecasting (under review)
- Designed **TAL-Net (Temporal Attention LSTM Network)** — bidirectional LSTM + temporal attention mechanism; trained on hourly US EIA grid data (California & Texas); achieved **0.48% MAPE** (demand) and **1.60% MAPE** (CO_2); outperformed 4 baselines (RNN, LSTM, RBM+MLP, RBM+LSTM); **19× improvement over ARIMA**

Feature Engineering, Interpretability & Scenario Analysis
- Engineered 27 input features (temporal indicators, lag variables 1h/24h/168h, rolling stats, fuel mix, weather); VIF analysis reduced 28→24 features without accuracy loss
- **SHAP analysis** confirmed coal (68.38%) and natural gas (23.99%) as dominant CO_2 drivers — physically consistent and audit-ready
- Solar penetration scenarios (10–50%): 50% penetration achieved **6.74% CO_2 reduction** (~127,000 car-equivalents annually); validated with paired t-tests, Bonferroni correction, bootstrap CI, Cohen's d — all $p < 0.0001$

Junior Data Analyst — Energy & Emissions Performance
Aug 2022 – Jan 2024

Navalt Solar and Electric Boats Pvt. Ltd

700+ Vessels processed	**200,000+** Rows / 200+ metrics	**85%** Efficiency gain (1 wk → 4 hrs)	**4 docs** Condensed into CII Guide

- Built **automated Python ETL pipeline** (pandas, NumPy) processing 200,000+ rows across 700+ vessels (200+ metrics) — weekly data reporting reduced from ~1 week to ~4 hours (**85% efficiency gain**)
- Performed **CII (Carbon Intensity Indicator) calculations** per IMO regulation (A–E rating: CO_2 ÷ DWT × distance) for full fleet; calculated SFOC/SFOS metrics for engine health monitoring and vessel performance benchmarking
- Prepared quarterly **ESI (Environmental Ship Index) compliance reports** for port authorities; maintained sea trial, drydock, and vessel performance databases
- Condensed 4 dense IMO regulatory documents into a single **CII Reference Guide**; built predictive models for fuel efficiency and CO_2 emissions using engine performance variables

RESEARCH & PUBLICATIONS

- **Two peer-reviewed journal papers co-authored** — AI-based electricity demand and CO_2 emissions forecasting using TAL-Net (Temporal Attention LSTM Network); currently under review
- Energy demand & CO_2 forecasting: 0.48% MAPE (demand), 1.60% MAPE (CO_2), 19× improvement over ARIMA; scenario modelling shows 6.74% CO_2 reduction at 50% solar penetration (~127,000 car-equivalents removed annually)
- Student volunteer — FICTA'25 (Frontiers of Intelligent Computing: Theory and Applications), international conference

SELECTED PROJECTS

Vehicle Emissions Power BI Dashboard — Interactive dashboard on CO_2 emissions across vehicle types (EU dataset); identified engine size as ~70% of emission variance with multi-dimensional manufacturer/country/fuel-type filters

Graduate Outcomes Analysis (Python) — Random Forest classification and K-means clustering on UK HESA data; identified four graduate profiles for targeted university support recommendations

Marketing Campaign Response Model (Python) — Logistic regression on 1,500 customer records; identified demographic features improving targeting accuracy by 20%

EDUCATION

MSc in Data Analytics — London Metropolitan University 2024 – 2025
B.Tech in Mechanical Automobile Engineering — SCT College of Engineering 2017 – 2021

KEY ACHIEVEMENTS

- **39.3% carbon footprint reduction** at 3T Additive Manufacturing through data-driven sustainability analysis and targeted decarbonisation interventions
- **Co-authored 2 peer-reviewed papers** — TAL-Net achieved 0.48% MAPE (demand), 1.60% MAPE (CO_2), 19× improvement over ARIMA ($p < 0.0001$)
- **£14.1M programme contribution** — delivered audited energy baseline (28,510 kWh / 6.65 tCO_2) for Innovate UK STRATA, validated against ND-NEED 2022 and CIBSE Guide F
- **85% efficiency gain** at Navalt — automated Python pipeline cut weekly data reporting from 1 week to 4 hours across 700+ vessels and 200,000+ data rows
- Progressed directly from intern to full WMG analyst role following performance at 3T Additive Manufacturing

CERTIFICATIONS

- Excel for Data Analytics — Simplilearn
- Statistics for Data Science — Simplilearn